PROJECT REPORT

CreditWise: Intelligent Loan Approval System

1. Introduction

In the modern financial sector, loan approval is a critical process that directly impacts both customer satisfaction and institutional profitability. Traditional manual evaluation methods are often time-consuming, inconsistent, and prone to human bias.

This project, **CreditWise Loan System**, aims to automate the loan approval process using Machine Learning techniques. The system analyzes applicant data and predicts whether a loan should be approved or rejected, ensuring faster, accurate, and unbiased decision-making.

2. Problem Statement

Secure Trust Bank faces challenges due to its manual loan approval process:

- Delay in decision-making
- Risk of rejecting eligible customers
- Approval of high-risk applicants

The objective is to build a **Machine Learning model** that:

- Learns patterns from historical data
- Predicts loan approval accurately
- Reduces human intervention and bias

3. Objectives

- To perform Exploratory Data Analysis (EDA)
- To preprocess and clean the dataset
- To build and compare multiple ML models
- To identify key factors affecting loan approval
- To develop an efficient prediction system

🔹 4. Dataset Description

The dataset contains applicant information such as:

- Income details
- Employment status
- Credit score
- Loan amount and term
- Financial indicators (Savings, DTI ratio)
- Target variable: **Loan Approved**

The dataset is synthetic and designed to simulate real-world scenarios.

🔹 5. Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and relationships:

- **Loan Approval Distribution:** Checked class balance
- **Income Distribution:** Right-skewed, presence of outliers
- **Credit Score Analysis:** Strong predictor of approval
- **DTI Ratio:** Lower ratio increases approval chances
- **Savings:** Higher savings linked to approvals
- **Correlation Heatmap:** Identified relationships between variables

Key Finding:
Credit Score is the most influential feature

🔹 6. Data Preprocessing

✓ Missing Value Handling

- Numerical → Mean imputation
- Categorical → Mode imputation

✓ Encoding

- Label Encoding for ordinal target variables

- One-Hot Encoding for nominal variables

✓ **Feature Scaling**

- Applied **StandardScaler**
- Ensures uniform feature range

✓ **Feature Selection**

- Removed irrelevant column: Applicant_ID

◆ 7. Model Development

The dataset was split into:

- **Training Set (80%)**
- **Testing Set (20%)**

Models Used:

- Logistic Regression
- KNN
- Naïve Bayes

◆ 8. Model Evaluation

Models were evaluated using:

- Accuracy Score
- Accuracy, Precision, Recall, F1-score

Performance Comparison

Model	Accuracy
Logistic Regression	~0.88
KNN	~0.78
Naïve Bayes	~0.86

🔹 9. Results

- Logistic Regression provided good baseline performance

Key Insight:

- Financial indicators like **Credit Score, Savings, and DTI Ratio** significantly influence approval

🔹 10. Feature Engineering

- Removed unnecessary columns
- Cleaned and structured dataset
- Improved model efficiency and accuracy

🔹 11. Conclusion

The CreditWise system successfully demonstrates how Machine Learning can automate loan approval decisions.

- Provides **fast and consistent predictions**
- Reduces human bias
- Improves decision accuracy

🔹 12. Future Scope

- Hyperparameter tuning for better performance
- Use advanced models (XGBoost, Neural Networks)
- Deploy as a web application
- Integrate real-time banking data

🔹 13. References

- Scikit-learn Documentation
- Python Libraries: Pandas, NumPy, Matplotlib, Seaborn and sklearn.
- Machine Learning concepts and online courses.